SeaElephant
for Mediterranean Port Cities



A practical platform for plastic reduction, education and the blue circular economy

One Earth One Ocean

Goal: Ridding waters worldwide of plastic waste

- Founded in 2011 with offices in Kiel and Munich, Germany
- Recognised non-for-profit status
- Over 400 members, permanent employees
- Funded through membership fees, donations and sponsorships



OEOO Activity Focus

- River and costal clean up facilities in 7 countries
- Environmental and marine conversation education and awareness
- Beach cleanups
- Ghost net recovery
- Research and development in micro-plastics







The plastic waste problem

GLOBAL SCALE

400 Mt
plastics produced globally in 2022[1]

9.5%
of plastics produced in 2022 came from recycled plastics[1]

19–23 Mt/year
plastic waste leaks into aquatic ecosystems[2]

MEDITERRANEAN URGENCY

730 t/day
plastic waste pollutes the Mediterranean[3]

95–100%
of floating litter in the Mediterranean is plastic[3]

EU IMPLEMENTATION GAP

42.1%
EU plastic packaging recycling rate in 2023[4]

55% by 2030
EU plastic packaging recycling target[5]



[1] Houssini et al., Communications Earth & Environment (2025): global plastics production in 2022; 38 Mt secondary plastics out of 400 Mt total.
[2] UNEP, Plastic Pollution page: 19–23 Mt/year leaks into aquatic ecosystems.
[3] UNEP/MAP, Pollution in the Mediterranean factsheet: 730 t/day; plastics are 95–100% of floating litter.
[4] Eurostat (2025): EU recycled 42.1% of plastic packaging waste in 2023.
[5] EUR-Lex, Packaging and packaging waste directive summary: 55% plastic packaging recycling target by 2030.

The problem is local, not just global



Global talks on plastic pollution are slow

Mediterranean port cities don’t need to wait for international consensus before acting.



Port cities are uniquely placed

At the intersection of tourism, trade, shipping, fishing, waste logistics and marine ecosystems.



The missing piece

A tangible platform that brings all local stakeholders together around practical action — not just targets.



“After 10 years savings roughly a million tons globally, now its time to scale. I invite you to join our mission, cleaning our oceans and rivers and overcome the barriers we saw in Geneva 2025.”

Günther Bonin,
Founder

What is SeaElephant?

A mobile blue circular-economy and education platform visiting Mediterranean port cities between 2028 and 2030.

> *How can our port city reduce unnecessary plastic, prevent waste from reaching the sea, improve circular material flows and build local capacity?*

Combines public education, professional capacity-building and practical next steps — tailored to each city.

- School programmes & citizen engagement
- Professional workshops for waste, port & city teams
- Technology demonstrations & material-quality learning
- Post-visit reporting & practical recommendations



Inside the SeaElephant

- 180m long repurposed bulk carrier designed for plastic recycling, research and training
- Onboard sorting, treating and returning of plastics to economic cycle as recyclable material
- Non-receycable waste converted into energy with onboard Waste to Energy plant under highest emission standards
- Ship also serves as an open laboratory and research centre, as well as an event and training venue



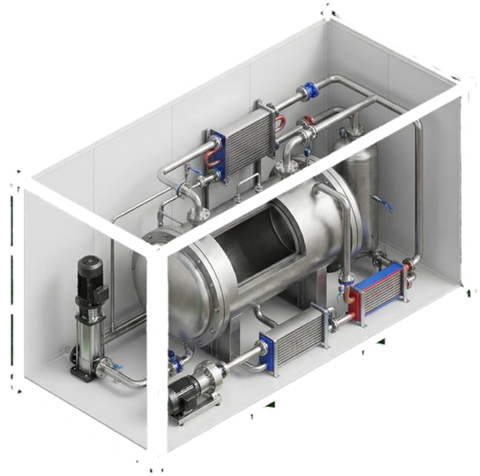
Example onboard technologies



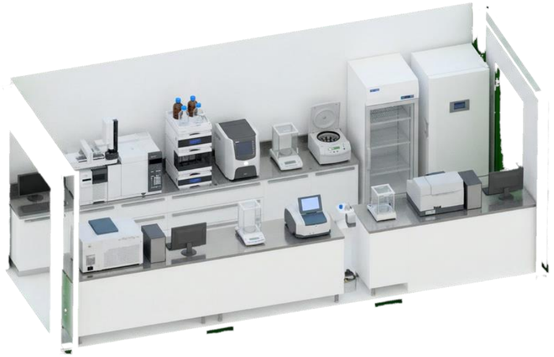
Cellulose Ester Unit — controlled chemical conversion and material processing



Biopolymer Unit — synthesis, separation and preparation of circular bio-based materials



eFuels / Kerosene Unit — thermal or catalytic conversion into fuel-related outputs



Analytical & Quality Control Lab — material testing, recyclate quality and process validation



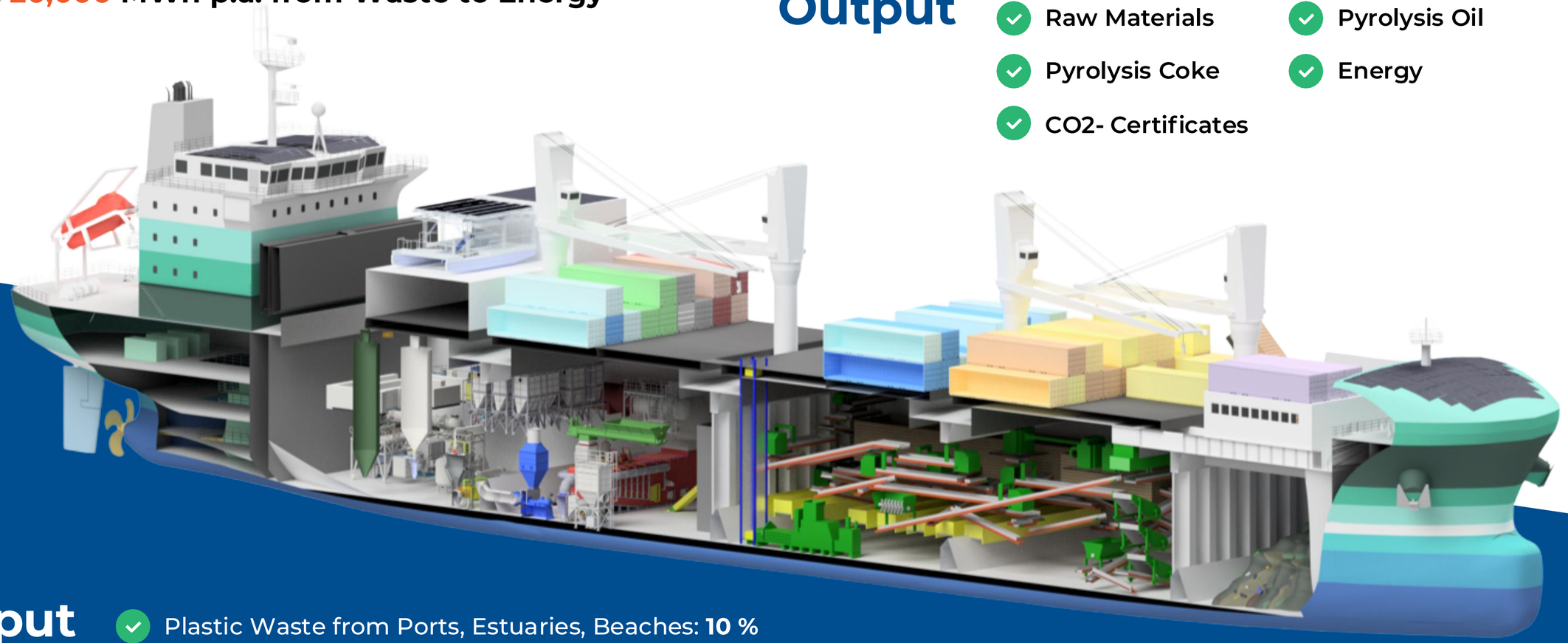
R&D / Process Development Lab — experimentation, optimisation and technology demonstration

A floating demonstrator

Processing capacity of Plastic Waste: 60,000 t/a Sorting, Baling, 20,000 MWh p.a. from Waste to Energy

Output

- Raw Materials
- Pyrolysis Oil
- Pyrolysis Coke
- Energy
- CO2- Certificates



Input

- Plastic Waste from Ports, Estuaries, Beaches: **10 %**
- Plastic Waste from Ships, Households and Industry: **90 %**

Imagine a day during a port call

Morning	Students and citizens step onto the ship to see how harbour plastics, tourism waste or fishing-related materials are collected, sorted, tested and explained.
Afternoon	Waste operators, port teams and waterfront businesses join practical workshops on recycling quality, prevention and circular blue-economy solutions.
Evening	The city hosts a public conversation on how local action can help protect the Mediterranean.

Three layers of impact

1

Public education & citizen engagement

School visits, guided tours, citizen science, awareness campaigns. Citizens see where their waste goes and why it matters.

2

Professional capacity-building

Workshops for waste companies, recyclers, port operators, municipal teams, universities, tourism businesses. Technology demonstrations and practical problem-solving.

3

Local learning & next steps

Impact reporting, waste-flow observations, stakeholder recommendations, pilot opportunities. What the city keeps after the ship leaves.

Why this matters for your city

- Show visible leadership on plastic reduction and Mediterranean protection
- Create a public moment citizens can see and understand
- Strengthen school and public education programmes
- Connect port sustainability with city environmental strategy
- Give local waste companies a platform as circular-economy partners
- Engage tourism and business sectors in practical sustainability
- Prepare for future circular infrastructure or Port 2.0 concepts

Working with your local waste ecosystem

SeaElephant works through the local system, not around it.

- Local waste companies, recyclers, utilities and port-waste operators are essential partners — they bring knowledge, infrastructure and relationships.
- A neutral platform: does not intervene in existing commercial or legal relationships.
- Creates a collaborative space for operators to engage with innovation, showcase expertise, train staff and strengthen public trust.
- Long-term value comes from what remains after the visit: better understanding, trained stakeholders, stronger partnerships and a clearer roadmap.

Who benefits and how

Stakeholder	What they gain
City leadership	Visible platform for circular economy and Mediterranean leadership
Port authority	Stronger sustainability positioning and harbour stewardship
Waste companies & recyclers	Innovation access, staff learning, public visibility as partners
Schools & young people	Hands-on learning linked to science, the sea and local action
Universities & researchers	Applied research, citizen science, education partnerships
Businesses & tourism	Practical sustainability engagement, cleaner waterfronts
Citizens	Understanding of how choices affect the sea
Funders & sponsors	Measurable outputs with high public visibility

What we’d ask from the host city

At this stage: explore feasibility together, not commit to a fixed plan.

1.	**Institutional support**	Signal that this is strategically relevant
2.	**Local coordination**	A lead contact to connect partners
3.	**Port & berth discussion**	Exploring practical options
4.	**Waste-ecosystem engagement**	Involving local operators
5.	**Education & stakeholder mobilisation**	Schools, universities, NGOs, businesses
6.	**Communications**	Joint visibility for the visit
7.	**Planning & permitting**	Navigating local requirements

Detailed operational planning, logistics and cost frameworks are developed separately.

A Mediterranean lighthouse network

Not just one visit — an invitation to help build a Mediterranean port-city movement.

Founding Lighthouse Cities Circle

Early cities shape the initiative, influence the Mediterranean manifesto, and are recognised as leaders — not just stops on a route.

Shared commitments:

- Protect the Mediterranean from plastic pollution
- Reduce unnecessary plastic use
- Improve reuse, sorting and recycling
- Educate citizens and young people
- Build capacity across local stakeholders
- Share lessons between port cities



SeaElephant helps Mediterranean port cities turn plastic-reduction ambition into practical blue circular-economy action — while connecting them in a shared effort to protect the Mediterranean.

A tangible platform to educate citizens, support schools, engage local waste companies, showcase new technologies, document challenges and build momentum toward cleaner seas.

Next steps

What resonates for your city?

Who else should be part of this conversation?

What would you need to explore feasibility for hosting?

Let’s start a conversation.